

strataq: measurement instruments for stochastic strategic systems

Susceptibility, reciprocity, and entropy production for quantal-response equilibria, calibrated on games with known answers

Sharath Sathish*

August 2026 — working draft v0.2 (grown at every closed gate)

Abstract

We present **strataq**, an open JAX library that treats logit quantal-response equilibrium (QRE) analysis as *instrumentation*: a susceptibility meter, a reciprocity meter, an entropy-production meter, and a phase locator, each an executable function with a stated confidence tier and a regenerable calibration artifact. The exact backbone is known (QRE = entropy-regularised response; Gibbs stationarity in potential games); the library’s contribution is (i) the *reciprocity transfer*: the observable equilibrium response $\chi^{\text{eq}} = (I - SB)^{-1}S$ is symmetric *iff* the normalised game has zero harmonic component, so strategic feedback neither creates nor destroys reciprocity—making the reciprocity defect $\mathcal{R}$ a payoff-free, λ -free *test* of potentiality; (ii) a separable Kronecker implementation of the Candogan decomposition giving the harmonic fraction α in near-linear time; (iii) exact-tier dynamics meters (currents, entropy production) on the joint-profile Glauber chain; and (iv) calibration of every meter at both ends of the α axis with payoffs known by construction: Rosenthal congestion on real Sioux Falls network data ($\mathcal{R} = 5.7 \times 10^{-17}$, Fisk KKT 7×10^{-15}) and Colonel Blotto ($\alpha = 0.69$, $\mathcal{R} = 0.12$). Pointing the calibrated instruments at seeded game families produced three findings we did not predict: $\mathcal{R}$ magnitudes scale with λ (only the zero test is λ -free); potential games *escape* criticality at high λ through equilibrium concentration, opening a supercritical wedge only for $\alpha \gtrsim 0.5$; and the response and dissipation layers decouple structurally in the near-harmonic regime—the correlation between response asymmetry and entropy production reverses sign at $\alpha = 0.95$, and no renormalisation of the response matrix repairs it. Every quantitative statement in this paper regenerates from fixed seeds (**make reproduce**); Appendix A maps each number to its artifact.

1 Introduction

Quantal response equilibrium [1] replaced exact best response with logit response, and in doing so imported the entire apparatus of Gibbs statistical mechanics into strategic analysis: the logit choice rule is the Gibbs variational principle, its log-partition function is a cumulant generating function, and in potential games [4] the induced Glauber dynamics are reversible with stationary measure $\pi \propto e^{\lambda\Phi}$ [5]. What the QRE literature has largely not done is treat this correspondence *instrumentally* — as licensing a set of measuring devices whose readings are meaningful whether or not they match a prediction.

*With agent-assisted implementation. Code, gate records, and every number’s regeneration path: <https://github.com/SharathSPhD/sage>.

This paper describes such a set of devices and what happened when we pointed them at things. The stance is deliberate: this is not a hypothesis-testing exercise. We built a susceptibility meter, a reciprocity meter, an entropy-production meter, and a phase locator; we calibrated each on systems where the correct reading is known analytically; and then we swept them across a one-parameter family of strategic environments — the harmonic fraction α of the Candogan decomposition [6] — and wrote down what they read. Three of the readings surprised us, and one of them refuted a correction we had proposed to our own instrument. All of it is reported, including the refutation.

Two disciplines make this more than a metaphor. First, every claim carries a confidence tier — *exact* (proved identity, implemented and verified to 10^{-12} – 10^{-16}), *derived* (proved in-house, numerically checked), *conjectured* (with a stated falsifier), or *speculative* — and the tier is visible in the code’s docstrings, the project’s claim ledger, and this paper. Second, every number below regenerates from a fixed seed by one command; Appendix A maps each to its machine-readable artifact, and each artifact was closed under an adversarial review whose objections and dispositions are part of the public record.

The library is **strataq**; the umbrella project is SAGE (Strategic Agent Game Engine). The research programme it instruments — non-equilibrium thermodynamics of strategic systems, with empirical estimation across pricing, energy, congestion and allocation domains — is described in the repository’s research documents; this paper is the toolkit paper, written first because the toolkit is, we believe, the most immediately useful artifact.

2 The instruments and their tiers

Notation: players $i \in \{1, \dots, N\}$ with finite action sets, mixed profiles σ_i , logit precision λ_i ; $C_i = \text{diag}(\sigma_i) - \sigma_i \sigma_i^\top$; $S = \text{blockdiag}(\lambda_i C_i)$; $B_{ij}(a, b) = \partial U_i(a) / \partial \sigma_j(b)$ the cross-payoff operator. All linear algebra involving $(I - SB)$ is performed on the tangent space $T = \bigoplus_i \{v : \mathbf{1}^\top v = 0\}$ via an explicit Helmert basis; C_i is rank-deficient by construction, and skipping this projection manufactures a spurious zero eigenvalue — a false criticality reading. Decomposition always acts on the *normalised* (effective) game: full externality symmetry is sufficient but not necessary for potentiality of the raw game [9].

2.1 The susceptibility meter (exact + derived)

The opponents-frozen response to an own-payoff field h_i is the static fluctuation–dissipation identity $\chi^{\text{part}} = \lambda_i C_i$ (exact; verified to 10^{-12} by autodiff of the log-partition). With strategic feedback, total differentiation of the fixed point gives the *strategic resolvent*

$$\chi^{\text{eq}} = \frac{d\sigma^*}{dh} = (I - SB)^{-1} S \quad \text{on } T, \quad (1)$$

likely folklore as comparative statics (we cite defensively) but load-bearing here: the same operator is the contraction certificate ($\rho(SB) < 1 \Rightarrow$ unique QRE), the bifurcation detector (eigenvalues of SB crossing 1), and the implicit-differentiation kernel (our custom VJP solves the transposed tangent system; its Jacobian reproduces (1) to machine epsilon). Verification: χ^{eq} matches central finite differences of the re-solved QRE to 1.3×10^{-8} over 50 random games.

2.2 The reciprocity meter (derived)

Result 1 (Reciprocity transfer). χ^{eq} is symmetric $\iff S(B - B^\top)S = 0 \iff$ (full support) $B = B^\top$ on $T \iff$ the normalised game has zero harmonic component.

The proof is three lines of resolvent algebra; the *content* is that strategic feedback neither creates nor destroys reciprocity, so the observable equilibrium response inherits the symmetry of the unobservable payoff operator exactly. A prior-art sweep (recorded in the repository, including the search scope) found no prior statement. Define the reciprocity defect $\mathcal{R} = \|\chi^{\text{eq}} - \chi^{\text{eq}\top}\|_F / \|\chi^{\text{eq}} + \chi^{\text{eq}\top}\|_F \in [0, \infty)$. Because Result 1 is a symmetry statement, *whether* $\mathcal{R} = 0$ is λ -free — nothing for a noise parameter to absorb, which is our lead response to the Haile et al. [3] non-falsifiability critique. Two properties discovered in calibration and now documented on the instrument: $\mathcal{R}$ is a norm ratio, unbounded above (matching pennies reads 1.2); and its *magnitude* scales with λ (approximately $\propto \lambda$ at small λ), so only the zero test is λ -robust.

2.3 The harmonic-fraction meter (exact)

The game graph is a Cartesian product of complete graphs; its Laplacian is a Kronecker sum with closed-form eigenspaces indexed by which coordinates vary. The Candogan projections therefore reduce to separable per-axis centering/averaging — an FFT-like transform, $O(\text{tensor size} \times N)$, no iterative solver — with the potential/harmonic split of each subset sector given by the degree-weighted mean across the players in the sector. We verify orthogonality (in the response-graph inner product), idempotence, exact reconstruction, and invariance of α under strategic equivalence, each to 10^{-9} or better. Families at exact target α are generated by mixing unit-norm potential and harmonic components — never by perturbing a Jacobian, whose skew part is coordinate-dependent [9].

2.4 The dissipation meter (exact)

Single-site logit revision on the joint profile space is a Glauber chain whose stationary distribution, probability currents J^* , and entropy production rate σ_{EP} we compute exactly (dense path, small games). In an exact potential game the revision probabilities are heat-bath conditionals of $\pi \propto e^{\lambda\Phi}$: detailed balance holds transition-by-transition (verified to 1.5×10^{-16}), $J^* = 0$, $\sigma_{\text{EP}} = 0$. Off potentiality the chain holds a non-equilibrium steady state that circulates and dissipates.

2.5 Solvers and the phase locator

Damped iteration, magnetic mirror descent [13] (last-iterate; agrees with damped to 7×10^{-13}), and pseudo-arclength continuation of the principal branch (agrees with Gambit’s homotopy [12] to 10^{-8} , and passes through the coordination pitchfork without stalling). The spectrum of SB on T supplies ρ , the distance to criticality $1 - \rho$, and bifurcation typing: $B = B^\top$ forces a real spectrum, so sustained cycles under logit dynamics cannot arise in full potential games — verified: the spectrum is exactly real on every potential game tested across $\lambda \in [0.5, 10]$, and visibly complex on harmonic games (cf. the logit limit cycles of Hommes and Ochea [15] at $\alpha = 1$).

3 Calibration: games with known answers

A meter is only as credible as the systems on which its reading is known. The α axis offers exactly two families with payoffs *free by construction* — no demand estimation contaminating the reading — and we anchored both ends before pointing the instruments anywhere else.

3.1 The $\alpha = 0$ anchor: congestion on real network data

Rosenthal congestion games [11] are exact potential games; in the population form, logit route choice *is* Fisk’s stochastic user equilibrium [10], whose convex potential (Beckmann integral plus scaled entropy) is known analytically. We implement the population engine (payoff field $F(x)$, potentiality tested by symmetry of DF — Sandholm’s externality symmetry [7], cited, not claimed) and solve SUE by damped Newton on the convex program in per-OD tangent coordinates; the damped logit-assignment map itself is *not* a contraction on steep-BPR networks (it two-cycles on the Braess diamond), a practical fact we document rather than hide.

Calibration readings (Table 1): on the Braess diamond, the autodiff gradient of the Beckmann integral equals the route-cost vector to 10^{-12} (an identity, and our sharpest end-to-end code check); on *real Sioux Falls data* (76 links, top-8 OD pairs, $k=3$ route sets from the TNTTP repository [17]), the Fisk KKT conditions hold to 7×10^{-15} , DF is symmetric to machine zero, and the toll-response reciprocity defect reads $\mathcal{R} = 5.7 \times 10^{-17}$. Link tolls enter route costs exactly linearly — the cleanest conjugate field in the programme — and the toll susceptibility matches finite differences to 2.8×10^{-9} , with the sign verified against actual re-solves. We state the scope once: route sets are k -shortest restrictions and the OD set is a subset, so computed link flows are *not* comparable to the repository’s best-known equilibrium flows (a diagnostic gap of 0.73 is reported as a diagnostic, never a certificate); the instrument identities are exact within the route set regardless.

3.2 The $\alpha > 0$ anchor: Colonel Blotto

Two-player Blotto over lattice-simplex allocation grids, budgets set by the experimenter (an exactly observable payoff perturbation). Readings: symmetric budget-3, three equal fields — $\alpha = 0.694$, $\mathcal{R} = 0.118$ at $\lambda = 1.5$; asymmetric field values — $\sigma_{EP} = 0.098$ with visible circulation. A degenerate instance (equal values, budget 2) has *constant* payoffs — every profile ties — and every meter reads exactly zero ($\sigma_{EP} = 5.4 \times 10^{-32}$): a correct null found while writing the positive test, kept as a regression. One calibration surprise worth naming: Blotto is *mixed*, not pure harmonic ($\alpha \approx 0.7$, not 1) — zero-sum does not imply harmonic-pure; RPS ($\alpha = 1$ exactly) remains the pure extreme.

4 Estimating λ : four routes and an honesty protocol

The rationality parameter is the most abused dial in the QRE literature — mis-specified demand, unmodelled heterogeneity and wrong payoff models all try to leak into it. Our defence is redundancy (unit `estimate.lambda`): four structurally different estimators run together, with their disagreement treated as a diagnostic rather than a nuisance.

1. **Frequency MLE** — multinomial likelihood of observed choice counts under $\sigma^*(\lambda)$, with a profile-likelihood confidence interval and a *flat-likelihood guard*: on any game whose QRE

Reading	Requirement	Measured	Artifact
$\nabla\psi = \lambda\sigma, \nabla^2\psi = \lambda^2C$	10^{-12}	holds	test suite
Gibbs $\pi \propto e^{\lambda\Phi}$ (congestion, $N=2,3$)	10^{-10}	2.5×10^{-16}	<code>gibbs_agreement</code>
$\sigma_{EP}, J^* $ on potential games	$< 10^{-12}$	holds	<code>equilibrium_reads_zero</code>
$\mathcal{R}$ on 5 exact potential games	$< 10^{-10}$	$\leq 9 \times 10^{-17}$	<code>reciprocity_potential</code>
$\mathcal{R}$ on RPS-3/RPS-5/matching pennies	> 0.1	0.69/0.43/1.20	<code>reciprocity_harmonic</code>
χ^{eq} vs finite differences (50 games)	10^{-6}	1.3×10^{-8}	<code>chi_fd_agreement</code>
Real spectrum of SB , potential games, $\lambda \leq 10$	rel. imag $< 10^{-8}$	0	<code>spectrum_reality</code>
Gambit agreement, 32 games $\times 2 \lambda$	10^{-8}	1.2×10^{-9}	<code>gambit_agreement</code>
Sioux Falls: $\mathcal{R}$ / DF defect / KKT	$10^{-10}/10^{-12}/10^{-8}$	$5.7 \times 10^{-17}/0/7 \times 10^{-15}$	<code>sioux_falls_calibration</code>
Blotto: $\alpha, \mathcal{R}$, degenerate null	see text	0.69, 0.12, 5×10^{-32}	<code>blotto_readings</code>

Table 1: Calibration summary. Every row regenerates from seed 20260808 via `make reproduce`; artifacts are JSON records under `benchmarks/results/` (Appendix A).

barely moves with λ (symmetric rock–paper–scissors is the extreme case, uniform at every λ), the estimator returns a warning, never a bare number.

2. **Autodiff MLE** — the same objective scored by JAX differentiation through an unrolled damped solve (convergence of the unroll verified to 10^{-11} at $\lambda = 20$); agreement with the grid MLE is a solver cross-check. Its CI is inherited from the profile likelihood and says so explicitly.
3. **χ moment-matching** — fit λ to an observed cross-response matrix; the route that works when choices are equilibrium-degenerate but responses are measurable (the pass-through setting of Section ??).
4. **Dispersion inversion** — invert mean choice entropy, with its own flat-entropy guard (a symmetric principal branch below its pitchfork is unidentified and says so).

Validation. On well-specified synthetic data (20 (game, λ^*) cases spanning an asymmetric anchor game and mixed- α family games, $\lambda^* \in [0.3, 6]$): median relative error 2.6% (MLE), 2.8% (dispersion), exact for χ -matching from oracle responses (artifacts `estimator_recovery.json`).

The diagnostic fires when it must. On data drawn from a λ -mixture (half 0.4, half 4.0 — not a QRE at any single λ) the estimator spread widens $\times 91$ over clean data and the protocol flags misspecification instead of averaging it away (`estimator_misspecification.json`). Two rules follow for all data work: report the whole family, never one number; and treat a flagged disagreement as a finding about the model.

First market contact. The same discipline held on real data (unit domains.electricity): asked to rationalise the July-2026 CAISO SP15 price dispersion, the domain’s stylised two-generator auction was *rejected* by its own one-moment fit — the model’s dispersion ceiling sits 40% below the spike-driven observation — and the pipeline reported the rejection rather than a λ (`electricity_lambda.json`). What *is* claimed from that data — a verified pair-level detailed-balance violation in the day-ahead series, and a genuinely at-null real-time series — belongs to the dissipation instruments and is developed in the companion paper [18].

From point estimates to posteriors. Unit `estimate.bayes` (ADR-0012) upgrades route 1 to a grid posterior over λ with calibrated frequentist coverage — and the calibration study is itself the best advertisement for the discipline: the pre-registered coverage floor FAILED on first run (34/50 at $\lambda^* = 1.8$), was chased to a real defect (a posterior spanning fewer than ~ 6 grid points quantises its credible interval to a couple of grid steps and undercovers at 78%), and the fix — a resolution guard that refuses to quote intervals from under-resolved grids, plus automatic refinement — restored 48/46/48 per λ^* , confirmed off the tested points by the red-team’s own probe. The scale fold $\sigma(\lambda, su) = \sigma(s\lambda, u)$ reappears here as an exact reparameterisation of the posterior (error 0.0): (λ, scale) is never identified from choices, and the API says so instead of hiding a ridge. The mixture diagnostic of the estimator family becomes a matched model comparison: an explicit two- λ mixture model beats the single- λ model decisively on mixture data (Bayes factor $> 10^2$) and is Occam-suppressed on clean data (0.11). All of it is callable by outsiders on their own data — `strataq.toolkit` and the public `/v1/toolkit` endpoints return these posteriors with their warnings attached.

5 Findings from pointing the instruments

With both anchors in place we swept seeded two-player 3×3 families of exact target α (mixed from unit-norm potential and harmonic components) and read every meter. Three findings follow, in the order discovered. All are rank-order statements at fixed $\lambda = 1.2$ on this family class unless noted; per-cell significance is neither claimed nor needed — surfaces and within-level correlations carry the content, with bootstrap intervals in the artifacts.

5.1 Finding 1: co-movement along α , with a marginal–conditional trap

Marginally, the meters co-move almost perfectly: Spearman $\rho(\mathcal{R}, \alpha) = 0.982$ ($n=2000$), $\rho(\sigma_{\text{EP}}, \alpha) = 0.990$ and $\rho(\sigma_{\text{EP}}, \mathcal{R}) = 0.993$ ($n=1000$). The last number is a trap: stratified by α , the within-level coupling $\rho(\sigma_{\text{EP}}, \mathcal{R} \mid \alpha)$ is $+0.80$ to $+0.88$ for $\alpha \leq 0.65$, degrades through $+0.61$ ($\alpha=0.75$) and $+0.32$ ($\alpha=0.85$), and *reverses* to -0.36 at $\alpha = 0.95$. The marginal 0.993 is α -driven confounding. (This stratification was demanded by the project’s adversarial reviewer; the marginal number alone had survived our own first write-up.)

5.2 Finding 2: the wedge, and criticality escape

The (λ, α) surface of $\rho(SB)$ shows two regimes (Fig. 1). On potential games, ρ is *non-monotone* in λ : it rises to ≈ 0.5 near $\lambda \approx 1.7$ and then collapses to zero, because the high- λ principal branch concentrates on a strict equilibrium and $C \rightarrow 0$ exponentially — $S = \lambda C \rightarrow 0$ despite the λ factor. Potential games *escape* criticality through equilibrium concentration. At $\alpha = 1$, mixed equilibria keep C bounded away from zero and ρ grows without bound. In between, a supercritical wedge opens: the median game first crosses $\rho = 1$ at $\alpha = 0.5$ ($\lambda \geq 8.5$); a 0.2 fraction of individual games crosses already at $\alpha = 0.4$; the frontier descends to $\lambda_c \approx 3$ by $\alpha = 0.8$. Inside the wedge the resolvent is near-singular and $\mathcal{R}$ magnitudes are direction-only — the surface artifact carries this caveat cell-by-cell.

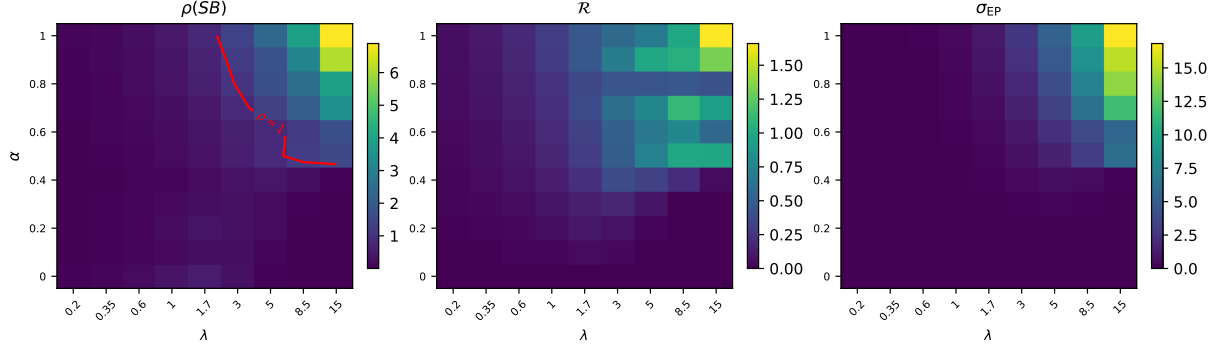


Figure 1: The (λ, α) surface: median $\rho(SB)$, $\mathcal{R}$, and σ_{EP} over seeded families (5 games/cell; α increases upward, λ rightward). The criticality escape at $\alpha = 0$ and the supercritical wedge are visible in the left panel. Regenerates from `experiments/phase_map.py`, seed 20260808.

5.3 Finding 3: the response and dissipation layers are distinct (a refuted repair, reported plainly)

To explain Finding 1 we hypothesised that $\mathcal{R}$'s denominator $\|\chi^{eq} + \chi^{eq\top}\|_F$ underflows at high α (the ratio measuring the residual symmetric sliver), and that the *numerator* $\|\chi^{eq} - \chi^{eq\top}\|_F$ would remain a valid circulation gauge. We tested both on the identical seeded games (paired design). Half held: $\rho(\mathcal{R}, 1/\text{den} \mid \alpha)$ climbs monotonically to 0.993 at $\alpha = 0.95$ — at high α , $\mathcal{R}$'s within-level variation *is* the shrinking symmetric response. But the repair failed: the numerator also decouples, $\rho(\text{num}, \sigma_{EP} \mid \alpha=0.95) = -0.37$ (95% bootstrap CI excluding zero), against $+0.77$ to $+0.86$ for $\alpha \leq 0.65$. No renormalisation of the response matrix recovers the dissipation ordering in the near-harmonic regime.

Our reading — argued, not proved — is that χ^{eq} is a local derivative at the QRE point (a marginal-strategy object) while σ_{EP} is a functional of the joint-profile stationary flux; nothing identifies one with the other except in potential games, where both vanish. The practical scope table this yields: $\mathcal{R}$ answers “*is this system potential?*” exactly, at every λ and every α ; σ_{EP} answers “*how hard is it circulating?*”; $\mathcal{R}$ levels are comparable across systems only for $\alpha \lesssim 0.65$ (measured at $\lambda = 1.2$, 3×3 , two players; the crossover's λ - and size-dependence is open). The refuted hypothesis, its post-hoc origin, and the sequence of events are recorded in the project's findings log; the gate protecting this result was specified after measurement as a regression contract, and says so on its face.

6 Software and reproducibility

`strataq` is a uv-workspace monorepo (github.com/SharathSPhD/sage) in JAX with float64 enforced at import — the susceptibility and dissipation computations are ill-conditioned near criticality and float32 is not acceptable there. Architecture in brief: three mathematical *engines* (finite strategic form; population games; Bayesian, deferred by recorded decision), with solvers, implicit differentiation and the dynamics/entropy machinery shared in an engine-agnostic core. Domains — congestion, Blotto, pricing, electricity — are *plugins*: exactly five declared objects (payoff oracle, action-grid builder, conjugate-field spec, loader, learn page) and zero core changes, enforced by a commit hook and CI. The conjugate-field declaration is load-bearing: a domain

that cannot name an observable payoff perturbation (e.g. penalty kicks) declares `NONE`, and the response instruments *refuse* to run for it rather than returning a meaningless number.

Reproducibility is contractual, not aspirational. A work unit closes only when its *gate* — code (tests, strict types, lint, coverage, no stubs), domain (the claim demonstrably true, with machine-readable artifacts), statistical (effect sizes with intervals, justified n , recorded seeds), documentation, and adversarial (a red-team review that sees the artifact and claim but never the implementation rationale) — is green, and every domain artifact must regenerate from its recorded seed via `make reproduce`. Anti-gaming rules are mechanical: deleting or `xfail`-ing a failing test blocks the commit; weakening a domain gate requires a recorded architecture decision; gate regressions fail CI. The adversarial layer earned its keep: of the roughly twenty objections raised across the units behind this paper, three were blocking defects in our own enforcement tooling, one exposed the marginal-conditional trap of Finding 1, and one forced the epistemic disclosures around Finding 3.

Performance notes for users: solvers are JIT-compiled (the 2,000-game α sweep with dissipation readings completes in ~ 20 s on CPU); the separable Hodge transform is near-linear in tensor size; the dense dynamics path is exact and intended for small games, with trajectory estimators (KLD, TUR bounds) planned for scale. `pygambit` is an optional oracle dependency, never a runtime requirement.

7 Related work and honest positioning

What is known, and cited as such. The entropy-regularised characterisation of logit response, the CGF structure of the log-partition, Gibbs stationarity of logit dynamics in potential games, and the symmetry-characterisation of potentiality are textbook [1, 5, 4, 7, 8, 14]. The intrinsic potential/harmonic decomposition is Candogan et al. [6]. An earlier draft of the research programme claimed the symmetry characterisation as new; the project’s own sanity check caught and downgraded it — that correction is part of the claim ledger’s public history, and we mention it as evidence the tier system binds us, not only others.

Nearest live work. Legacci et al. [9] prove, in Shahshahani geometry, that harmonic games are incompressible under continuous-time exponential weights — volume preservation, constants of motion, Poincaré recurrence — and leave discrete-time behaviour and mixed games open. Our programme sits at a different layer: discrete logit QRE, the observable equilibrium-response matrix, and non-equilibrium observables (currents, dissipation), with calibration on real data. Their precision fix (potentiality of the *effective* game; Lemma C.2) is baked into our decomposition path. We found no prior statement of the reciprocity transfer (Result 1) or of response-matrix asymmetry as an observable proxy for harmonic content; the sweep and its scope are recorded in the repository’s literature file.

What this paper does not claim. $\mathcal{R}$ is not a universal circulation gauge (Finding 3 refutes exactly that); λ estimated from data absorbs unmodelled heterogeneity, and low λ is not proof of irrationality [3]; the Sioux Falls exercise is calibration on computed equilibria, not an empirical claim about drivers; and the thermodynamic language is exact only where the identities are exact — for non-potential games what survives is the NESS framing, which is where the interesting readings live. Empirical contact (scanner panels, electricity offer curves, λ estimator families and conduct testing) is the subject of the programme’s later papers; the closest empirical precedent for the dissipation angle is Xu and Wang [16], who identify an Edgeworth cycle via nonzero entropy production in experimental market data.

A Every number’s provenance

Every quantitative statement in this paper maps to a machine-readable artifact in `benchmarks/results/` (JSON, schema `strataq_bench.BenchmarkResult`: metrics, effect sizes with intervals, seed, config reference, library version, timestamp). All regenerate with `make reproduce` (seed 20260808 unless stated). The gate records in `gates/*.yaml` carry, for each unit, the claim, the section-by-section pass state, and the adversarial reviewer’s objections with their dispositions — including the two rounds on Finding 3’s unit, whose gate discloses on its face that its thresholds were specified post-measurement as a regression contract.

Statement in text	Artifact
Calibration table (Table 1)	the ten artifacts named in its rows
$\rho(\mathcal{R}, \alpha) = 0.982$, $n=2000$	<code>reciprocity_alpha_sweep.json</code>
$\rho(\sigma_{EP}, \alpha) = 0.990$; $\rho(\sigma_{EP}, \mathcal{R}) = 0.993$; within-level reversal	<code>chain_comovement.json</code>
Phase surface; wedge onset; criticality escape	<code>phase_map.json</code> , <code>phase_map_surface.json</code>
Finding 3 (H2 confirmed, H1 refuted, CIs)	<code>decoupling_mechanism.json</code>
Solver agreements (damped/mirror/Gambit/implicit/branch)	<code>solver_cross_agreement.json</code> , <code>gambit_agreement.json</code> , <code>implicit_chi_agreement.json</code> , <code>branch_agreement.json</code>
Sioux Falls calibration; UE diagnostic honesty note	<code>sioux_falls_calibration.json</code>
Engine-2 identities (Beckmann gradient, KKT, DF , toll- χ)	<code>population_identities.json</code>
Blotto readings incl. the degenerate null	<code>blotto_readings.json</code>

The findings log (`memory/findings.md`, entries F-0001–F-0007) records each anomaly with the config that produced it, its chase status, and its resolution — including the refuted hypothesis of Finding 3 and the provenance of its post-hoc origin. The claim ledger (`memory/claims.md`) carries every scientific claim’s confidence tier and history, including downgrades.

References

- [1] McKelvey, R.D., Palfrey, T.R.: Quantal response equilibria for normal form games. *Games and Economic Behavior* 10(1), 6–38 (1995).
- [2] Goeree, J.K., Holt, C.A., Palfrey, T.R.: *Quantal Response Equilibrium: A Stochastic Theory of Games*. Princeton (2016).
- [3] Haile, P.A., Hortaçsu, A., Kosenok, G.: On the empirical content of quantal response equilibrium. *American Economic Review* 98(1), 180–200 (2008).
- [4] Monderer, D., Shapley, L.S.: Potential games. *Games and Economic Behavior* 14(1), 124–143 (1996).
- [5] Blume, L.E.: The statistical mechanics of strategic interaction. *Games and Economic Behavior* 5(3), 387–424 (1993).
- [6] Candogan, O., Menache, I., Ozdaglar, A., Parrilo, P.A.: Flows and decompositions of games: harmonic and potential games. *Mathematics of Operations Research* 36(3), 474–503 (2011).
- [7] Sandholm, W.H.: *Population Games and Evolutionary Dynamics*. MIT Press (2010).

- [8] Balduzzi, D., et al.: The mechanics of n -player differentiable games. *ICML* (2018).
- [9] Legacci, D., Mertikopoulos, P., Pradelski, B.S.A.: A geometric decomposition of finite games: convergence vs. recurrence under exponential weights. *ICML* (2024). arXiv:2405.07224.
- [10] Fisk, C.: Some developments in equilibrium traffic assignment. *Transportation Research B* 14(3), 243–255 (1980).
- [11] Rosenthal, R.W.: A class of games possessing pure-strategy Nash equilibria. *Int. J. Game Theory* 2, 65–67 (1973).
- [12] Turocy, T.L.: A dynamic homotopy interpretation of the logistic quantal response equilibrium correspondence. *Games and Economic Behavior* 51(2), 243–263 (2005).
- [13] Sokota, S., et al.: A unified approach to reinforcement learning, quantal response equilibria, and two-player zero-sum games. *ICLR* (2023).
- [14] Matějka, F., McKay, A.: Rational inattention to discrete choices. *American Economic Review* 105(1), 272–298 (2015).
- [15] Hommes, C.H., Ochea, M.I.: Multiple equilibria and limit cycles in evolutionary games with logit dynamics. *Games and Economic Behavior* 74(1), 434–441 (2012).
- [16] Xu, B., Wang, Z.: Measurement and application of entropy production rate in human subject social interaction systems. arXiv:1107.6043 (2011).
- [17] Stabler, B., Bar-Gera, H., Sall, E.: Transportation networks for research. github.com/bstabler/TransportationNetworks.
- [18] S. Sathish. The non-equilibrium thermodynamics of quantal response. Working draft, 2026. In the same repository.